Submission PHYS1-FINAL-SUB02

Question PHYS1-FINAL-Q01
work / setup:
1. $v=v\blacksquare+at=8+2(5)=18$ m/s
crossed-out arithmetic: [smudged]
FINAL (maybe): $v=18$ m/s; displacement=65 m plus 1

Question PHYS1-FINAL-Q02
work / setup:
1. $N=mg=19.6$ N; friction=$\mu N=3.9$ N
2. $\Sigma F=13-3.9=9.1$ N
crossed-out arithmetic: [smudged]
FINAL (maybe): $a=4.54$ m/s²; friction opposes motion plus 1

Question PHYS1-FINAL-Q03
work / setup:
1. $mgh=\frac{1}{2}mv^2 \Rightarrow v=\sqrt{(2gh)}=\sqrt{(2\cdot 9.8\cdot 6)}$
crossed-out arithmetic: [smudged]
FINAL (maybe): $v=10.84$ m/s plus 1

Question PHYS1-FINAL-Q04
work / setup:
1. p_total,before = p_total,after
crossed-out arithmetic: [smudged]
FINAL (maybe): total momentum before equals after when external impulse is negligible plus 1

Question PHYS1-FINAL-Q05
work / setup:
1. Impulse is area under F(t): $J=F\Delta t=6(0.5)=3$ N·s.
crossed-out arithmetic: [smudged]
FINAL (maybe): impulse=3 N·s; momentum changes by 3 kg·m/s plus 1

Question PHYS1-FINAL-Q06
work / setup:
1. $v=v\blacksquare+at=5+2(5)=15$ m/s
crossed-out arithmetic: [smudged]
FINAL (maybe): $v=15$ m/s; displacement=50 m plus 1

Question PHYS1-FINAL-Q07
work / setup:
1. $N=mg=68.6$ N; friction=$\mu N=13.7$ N
2. $\Sigma F=23-13.7=9.3$ N
crossed-out arithmetic: [smudged]

FINAL (maybe): $a=1.33$ m/s^2; friction opposes motion plus 1

Question PHYS1-FINAL-Q08

work / setup:

1. $mgh=\frac{1}{2}mv^2 \Rightarrow v=\sqrt{(2gh)}=\sqrt{(2\cdot9.8\cdot5)}$

crossed-out arithmetic: [smudged]

FINAL (maybe): $v=9.90$ m/s plus 1

Question PHYS1-FINAL-Q09

work / setup:

1. p_total,before = p_total,after

crossed-out arithmetic: [smudged]

FINAL (maybe): total momentum before equals after when external impulse is negligible plus 1

Question PHYS1-FINAL-Q10

work / setup:

1. Impulse is area under F(t): $J=F\Delta t=6(0.5)=3$ N·s.

crossed-out arithmetic: [smudged]

FINAL (maybe): impulse=3 N·s; momentum changes by 3 kg·m/s plus 1